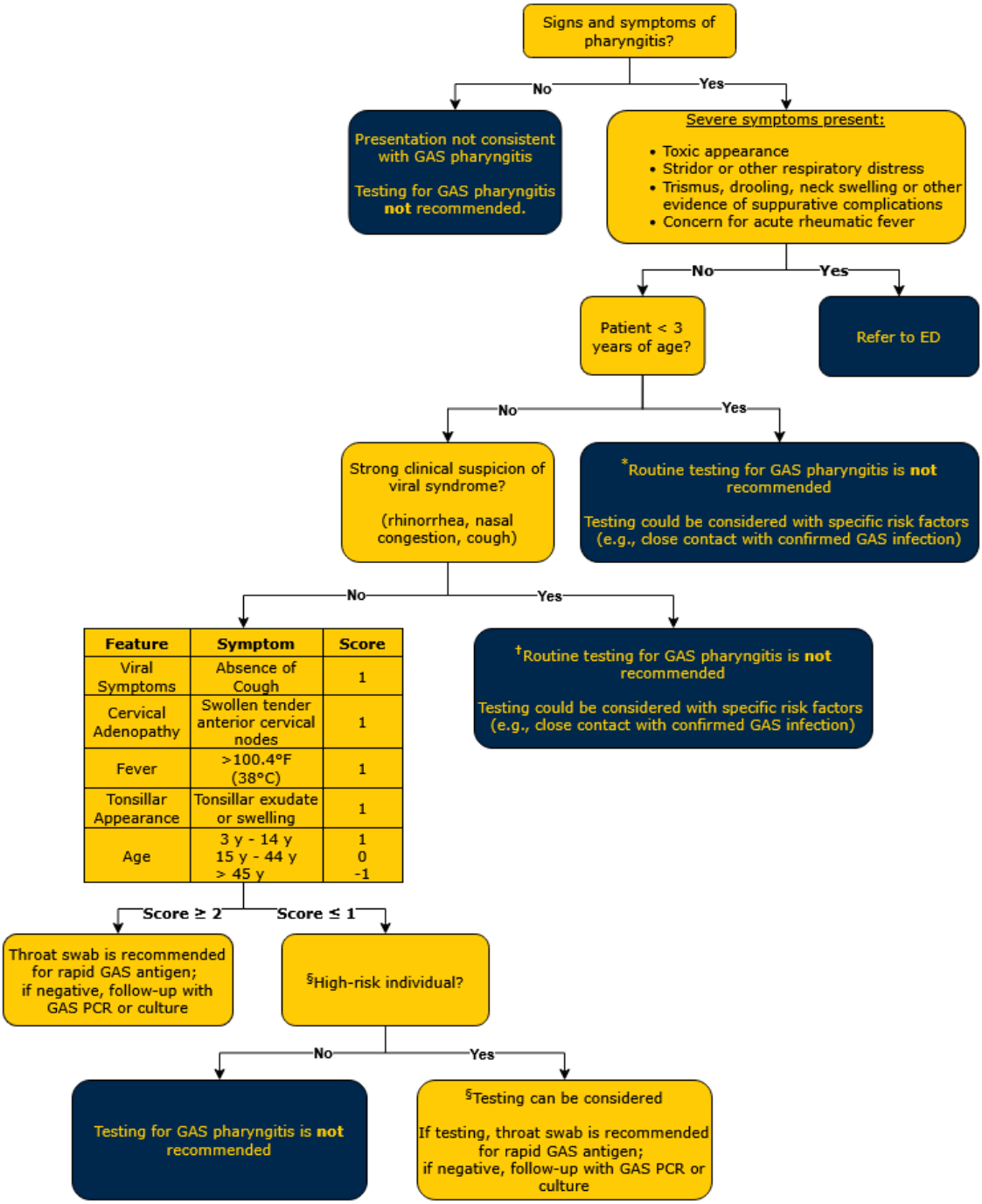
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# Antibiotic Treatment Guidelines for Group A Streptococcal Pharyngitis in Pediatric Patients

This guideline is designed to provide guidance for pharyngitis caused by group A Streptococcus (GAS) in most children. Patients with a recent history of oropharyngeal or other deep neck space surgery or infection are excluded. Patients who have evidence of suppurative complications on exam (such as peritonsillar abscess) OR patients with suspected acute rheumatic fever will need additional management that is outside the scope of this guideline.




When to refer to ENT: 7 episodes of bacterial pharyngitis in a year, or 5 per year in 2 years, or 3 per year in 3 years.

When to refer to ID: Can consider referral if there is concern for carrier status and parents wish to discuss this further.

*Testing in patients < 3 years of age is not recommended due to rarity of true GAS pharyngitis and acute rheumatic fever.

†Testing for GAS in patients with concomitant symptoms of viral illness is not recommended as a positive result if of questionable diagnostic value. Rates of asymptomatic carriage of GAS are reported as high as 25% (AAP Red Book). In these cases risk of transmission and suppurative/nonsuppurative complications is low.

§High-risk individuals include: those with sore throat and household exposure to GAS (e.g., living or sleeping in the same indoor shared space as a person diagnosed with GAS infection), a history of a previous rheumatic fever diagnosis, or symptoms or signs suggestive of complicated local or systemic GAS infection (e.g., peritonsillar or retropharyngeal abscess, scarlet fever and/or toxic shock syndrome).

| Recommended Therapy | Duration/Comments |
|---|---|
| 1st line:<br>**Penicillin V**<br>*Preferred*:<br>**5 days:** 50 to 75 mg/kg/day in divided doses QID (max: 3.2 g/**day)**<br>*Alternative for risk factors (see comments):*<br>**10 days**:<br>≤ 27 kg: 250 mg BID<br>> 27 kg: 500 mg BID<br><br>OR<br><br>**Amoxicillin***<br>*Preferred:*<br>**5 days:** 45 mg/kg/DOSE PO BID (max: 1 g/dose)<br>*Alternative for risk factors (see comments):*<br>**10 days:** 50 mg/kg/DOSE PO once daily (max: 1 g/dose)<br><br>OR<br><br>**Penicillin G benzathine**<br>≤ 27 kg: IM: 600,000 units as a **single dose**<br>> 27 kg: IM: 1,200,000 units (1.2 million units) as a **single dose**<br><br>Low-risk penicillin allergy:[1]<br>**Cephalexin** 20 mg/kg/DOSE PO BID (max: 500 mg/dose) for **10 days**<br><br>High-risk penicillin allergy or low-risk allergy and inability to take cephalexin:[1]<br>**Clindamycin** 7 mg/kg/DOSE PO TID (max: 300 mg/dose) for **10 days**<br>OR<br>**Azithromycin** 12 mg/kg/DOSE PO once daily (max: 500 mg/dose) for **5 days** | **Duration of amoxicillin and penicillin**<br>For most children, we recommend shared decision making with families with a preference for **5 days** versus **10 days**.<br><br>**5 days** is recommended for the following reasons:<br>• Symptoms are typically resolved by 5 days regardless of treatment<br>• Longer courses do not affect likelihood of suppurative complications<br>• Risk of acute rheumatic fever (ARF) is very low in the US for most populations (see more details below)<br>• Transmission risk significantly decreases after just one dose of antibiotics<br>• Longer courses are associated with increased adverse drug events and antimicrobial resistance<br><br>**10 days** (5 days for azithromycin) is recommended for the following scenarios:<br>• If using alternative agents due to allergy (cephalexin or clindamycin) or<br>• If there is concern for current or past ARF or<br>• Concern for poststreptococcal reactive arthritis or glomerulonephritis or<br>• Strong desire for bacterial eradication, such as:<br>  ○ local outbreak of ARF, poststreptococcal glomerulonephritis, or severe or invasive GAS disease<br>  ○ an outbreak of GAS pharyngitis in a closed or semi-closed community<br>  ○ family history of ARF<br>  ○ multiple episodes of documented symptomatic GAS pharyngitis occurring within a family for many weeks despite appropriate therapy<br>  ○ when tonsillectomy is being considered solely because of frequent GAS isolations<br>• Indigenous or Polynesian ancestry including Hawaii and American Samoa, indigenous peoples of New Zealand and Australia or other Pacific Islands<br>• Recent travel to **areas with high-risk of ARF which include parts of**:<br>  ○ **Africa**<br>  ○ **Asia**<br>  ○ **Pacific**<br><br>The **United States**, Canada, and most of Europe are considered **low-risk populations.** The incidence in the United States is unknown but is believed to be < 0.5%.<br><br>**Comments**<br>• GAS is uniformly susceptible to all beta-lactams (penicillins and cephalosporins), whereas macrolide and clindamycin resistance rates of close to 30% have been reported among invasive GAS isolates in the US. These agents should only be used in the case of true beta-lactam allergies.<br>• In case of penicillin allergy, review β-lactam allergy evaluation and empiric guidance and remove allergy label if possible. If not, consider referral to Allergy clinic to provide guidance on future episodes. |

* Renal adjustment may be necessary. See Pediatric Antimicrobial Dosing Guidelines.

[1] Low-risk allergies include: pruritus without rash, remote (> 10 years) unknown reaction, mild to severe rash with no other symptoms; see β-lactam allergy evaluation and empiric guidance for further information.

[2] High-risk allergies include: anaphylaxis, respiratory symptoms (chest tightness, bronchospasm, wheezing, cough), angioedema (swelling, throat tightness), cardiovascular symptoms (hypotension, dizzy/lightheadedness, syncope/passing out, arrhythmia); see β-lactam allergy evaluation and empiric guidance for further information.

[3] Non-IgE mediated allergies are contraindications to further beta-lactam use unless approved by Allergy: organ damage (kidney, liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, rash with mucosal lesions (Stevens Johnson Syndrome/Toxic Epidermal Necrosis), rash with pustules (acute generalized exanthematous pustulosis), rash with eosinophils and organ injury (DRESS – drug rash eosinophilia and systemic symptoms), rash with joint pain, fever, and myalgia (Serum Sickness); see β-lactam allergy evaluation and empiric guidance for further information.

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| | Citation |
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**Revision History**

| Date | Revision |
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*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.*